

QVRF

A Verifiable Random Function from Standardized Post Quantum Primitives, with Per Block Randomness
Extracted from Byzantine Finality

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Released for public cryptanalysis. Proven results and conjectures are distinguished throughout.

Abstract

The advent of scalable quantum computation dissolves the number theoretic foundations on which contemporary verifiable randomness is built. Shor's algorithm reduces both integer factorization and the discrete logarithm to polynomial time, and with them the elliptic curve verifiable random function of RFC 9381, whose secret key a quantum adversary recovers outright. We present QVRF, a verifiable random function whose security rests exclusively on primitives standardized by the United States National Institute of Standards and Technology: the SHA-3 family (FIPS 202) and the module lattice signature ML-DSA (FIPS 204), with a stateless hash based option from SLH-DSA (FIPS 205). QVRF comprises three constructions. A per input verifiable random function derives its image from a derandomized ML-DSA signature and a transparent hash based argument of correct evaluation; we establish unique provability under strong existential unforgeability, and pseudorandomness in the quantum random oracle model. A per block beacon extracts protocol randomness from the Byzantine fault tolerant finality certificate that consensus already produces, and we prove by explicit reduction that any adversary biasing the beacon yields an adversary breaching consensus safety, so beacon unpredictability inherits the safety of the ledger at the cost of a single hash evaluation and no additional protocol message. We then address a gap that the first construction cannot close. Uniqueness there presumes a signer that derandomizes, and a FIPS 204 signature carries no randomizer, so no verifier can check that presumption; where the drawing party is itself the adversary, as in stake weighted committee sortition, the presumption is unavailable. For that setting we give a third construction whose uniqueness is structural rather than assumed, drawing from committed one time preimages under SHA-3 alone, for which re drawing is impossible by construction and the cost is one hash and one Merkle opening. We give the security games, the reductions with concrete advantage bounds, a quantitative analysis of the work factors an adversary must expend under classical and quantum attack, and an explicit account of what is proven against what is assumed.

Keywords: *verifiable random function, post quantum cryptography, lattice cryptography, quantum random oracle model, Byzantine fault tolerance, randomness beacon, Shor's algorithm, Grover's algorithm.*

1 Introduction

A verifiable random function, introduced by Micali, Rabin and Vadhan, is a keyed function whose outputs are simultaneously unique, pseudorandom, and publicly verifiable against a committed public key. In a distributed ledger the primitive is load bearing rather than peripheral: leader election, committee sampling, and shard assignment are decided by its output, so an adversary able to predict or bias that output can bias the protocol that consumes it. The standardized instantiation, the elliptic curve verifiable random function of RFC 9381, derives its guarantees from the hardness of the elliptic curve discrete logarithm problem.

That foundation does not survive the quantum transition. Shor's algorithm solves the discrete logarithm in quantum polynomial time, collapsing the work factor of an elliptic curve key recovery from the classical square root of the group order to a polynomial in the bit length. A cryptographically relevant quantum computer therefore voids every RFC 9381 verifiable random function and, with it, the integrity of the randomness on which the consuming consensus depends. Shor's algorithm defeats factorization, discrete logarithm, and pairings alike, whereas the hash and lattice assumptions on which QVRF is built suffer only Grover's quadratic loss, which a doubled parameter absorbs.

Randomness sits at the root of leader selection, so a predictable beacon is a corruptible chain. Post quantum verifiable random functions have been proposed from lattice assumptions, yet two obstacles stand between that work and deployment. First, several constructions rest on assumptions NIST has not standardized, weakening the long term assurance that institutional and sovereign adopters require. Second, the prevailing design treats the function as a self contained object that a party evaluates, proves, and broadcasts, adding a computation, an argument, and a message to a path on which latency and bandwidth are scarce.

This paper resolves both obstacles. Our contributions are the following.

- A per input verifiable random function (Section 4) from derandomized ML-DSA and a hash based argument of correct evaluation, with unique provability reduced to strong existential unforgeability and pseudorandomness reduced to unforgeability in the quantum random oracle model.

- A per block beacon (Section 5) obtained by extracting from the Byzantine finality certificate the protocol already produces, with an explicit reduction from beacon bias to a consensus safety violation. The construction adds one hash evaluation per block and no protocol message.
- A sortition construction (Section 6) drawing from committed one time preimages, whose uniqueness is structural rather than assumed, for the setting in which the drawing party is the adversary and no assumption about its signing behaviour is available.
- A quantitative security analysis (Section 7) giving the classical and quantum work factors an adversary must expend against each component, and the parameter choices that hold each above the 128 bit quantum floor.

We are precise about the novelty claimed. We claim composition entirely from standardized primitives, deployment in a production ledger, and the finality extraction with its bias resistance reduction. We do not claim priority over the abstract notion of a post quantum verifiable random function, which precedes this work.

2 Preliminaries

Notation. We write $x \parallel y$ for concatenation, $|x|$ for byte length, and $x \leftarrow \$ S$ for uniform sampling from a finite set. The security parameter is n . We target 128 bits of classical security and account for Grover's quadratic speedup explicitly in Section 6.

2.1 Hash functions

Let H denote SHA3-256 and XOF denote SHAKE256 with a 256 bit output unless otherwise stated. Throughout the analysis XOF is modelled as a random oracle to which the adversary has quantum superposition access, that is, the quantum random oracle model of Boneh, Dagdelen, Fischlin, Lehmann, Schaffner and Zhandry. We write q for the number of quantum oracle queries.

2.2 Signatures and the determinism requirement

Let $\Sigma = (\text{KeyGen}, \text{Sign}, \text{Vf})$ be ML-DSA (FIPS 204) in its derandomized mode, in which $\text{Sign}(\text{sk}, m)$ is a deterministic function of key and message. We require strong existential unforgeability under chosen message attack: no efficient adversary with a signing oracle outputs a valid pair (m, σ) it did not receive, even for a queried message. The strong variant, rather than plain existential unforgeability, is what unique provability demands, since it must hold even for an adversarially generated public key where a second verifying signature on a queried message would otherwise be admissible. The conservative variant substitutes SLH-DSA (FIPS 205), whose security rests on H alone.

2.3 Argument system

Let $(\text{Prove}, \text{Vfy})$ be a transparent, hash based, non interactive argument of knowledge, instantiated by a STARK, with completeness and knowledge soundness in the quantum random oracle model reducing to the collision resistance of H . We treat it modularly; its soundness error $\varepsilon_{\text{STARK}}$ is drawn from the prover's specification.

2.4 Adversary and network model

We consider a quantum polynomial time adversary with quantum access to the random oracle and classical access to every protocol interface. For the beacon we adopt the standard partially synchronous Byzantine model: after global stabilization the adversary schedules delivery within a known bound, corrupts adaptively up to a strictly sub one third weighted threshold, and is rushing. An honest validator never signs two conflicting blocks; one that does is faulty and counted in the adversary's budget. The per input construction requires no network assumption. The design principle of Section 5 is that the beacon introduces nothing beyond the model in which the underlying protocol already proves safety.

3 Security definitions

A verifiable random function is a tuple $(\text{Gen}, \text{Eval}, \text{Prove}, \text{Vfy})$ with $\text{Gen}(1^n) \rightarrow (\text{pk}, \text{sk})$, $\text{Eval}(\text{sk}, x) \rightarrow y$, $\text{Prove}(\text{sk}, x) \rightarrow \pi$, and $\text{Vfy}(\text{pk}, x, y, \pi) \rightarrow \{0,1\}$. We state the three properties as games to fix the adversary's interface.

3.1 Unique provability

For every pk , including one chosen adversarially, and every x , no two tuples $(y, \pi), (y', \pi')$ with $y \neq y'$ both verify. The adversary wins Uniq by producing such a collision for any pk of its choice. This is the property that denies a party the ability to grind candidate outputs and publish the one it prefers.

3.2 Pseudorandomness

Game PR , quantum random oracle model. The challenger runs Gen , gives pk to A , and answers Eval and Prove queries. A submits an unqueried challenge x^* ; the challenger flips b and returns $y^* = \text{Eval}(\text{sk}, x^*)$ if $b = 0$, else $y^* \leftarrow \$ \{0,1\}^{256}$. A continues, may not query x^* , and outputs b' . The advantage is

$$\text{Adv}_{\text{PR}}(A) = \left| \Pr[b' = b] - 1/2 \right|,$$

required negligible in n .

3.3 Verifiability

Completeness: honest tuples verify with probability one. Soundness: no efficient A produces (x, y, π) with $V_{fy} = 1$ and $y \neq \text{Eval}(\text{sk}, x)$.

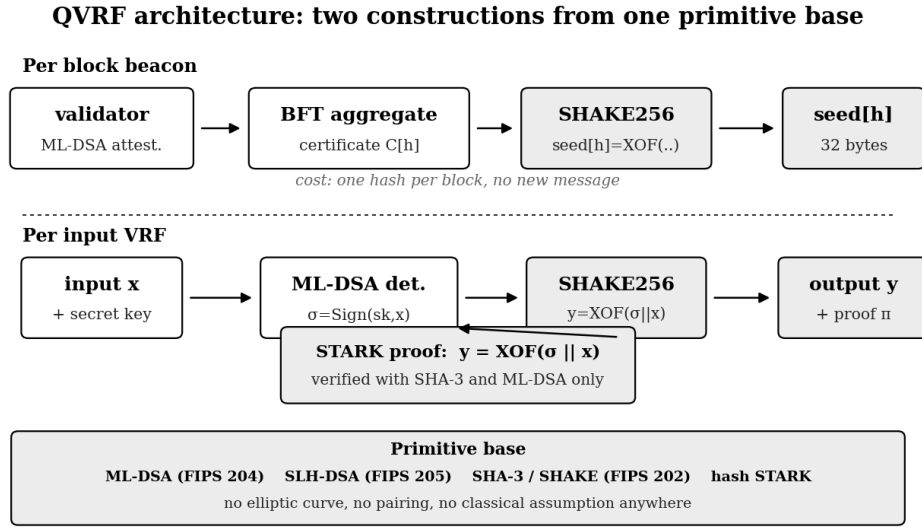


Figure 1. The two QVRF constructions and their shared primitive base. The beacon lane (top) reads the finality certificate consensus already emits and hashes it once. The per input lane (bottom) derives an image from a deterministic signature and certifies the derivation with a STARK. Neither lane touches an elliptic curve or a pairing.

4 The per input construction

4.1 Algorithms

```

Gen(1^n):      (pk, sk) <- Σ.KeyGen()
Eval(sk, x):   σ <- Σ.Sign_det(sk, x);   y <- XOF(σ || x, 256)
Prove(sk, x):  σ <- Σ.Sign_det(sk, x);
Vfy(pk, x, y, π):  π_s <- Prove[ y = XOF(σ || x) ];   return (σ, π_s)
                return Σ.Vf(pk, x, σ) AND Vfy[π_s : y = XOF(σ || x)]
    
```

The argument transmits the signature so the verifier fixes the unique σ for (pk, x) , and the STARK certifies y was derived from that σ by the prescribed function. Verification invokes only ML-DSA verification and SHA-3.

4.2 Unique provability

Theorem 1. If Σ is strongly existentially unforgeable and $(\text{Prove}, \text{Vfy})$ has knowledge soundness error ϵ_{STARK} , then for every adversary A against Uniq there is an adversary B , of essentially A 's running time, with $\text{Adv}_{\text{Uniq}}(A) \leq \text{Adv}_{\text{SUF-CMA}}(B) + \epsilon_{\text{STARK}}$.

Proof. Let A output (pk, x, y, π) and (pk, x, y', π') with $y \neq y'$, both accepted. Each argument parses as (σ, π_s) with $\Sigma.Vf(pk, x, \sigma) = 1$ and a proof that the image equals $\text{XOF}(\sigma || x)$. If the two carry the same σ , then $\text{XOF}(\sigma || x)$ is one determined value, forcing $y = y'$ unless a π_s violates soundness, of probability at most ϵ_{STARK} . If they carry $\sigma \neq \sigma'$, both verify on x under pk , and B returns the second as a strong forgery. Summation gives the bound. ■

The reliance on derandomized signing, so that the honest σ is itself unique, is the crux and the point most easily missed: under hedged ML-DSA a message admits many signatures, the second case ceases to be a forgery, and unique provability fails. It must be stated plainly that this is an assumption about the signer and not a rule a verifier can enforce. A FIPS 204 signature does not carry its randomizer, so a derandomized signature and a hedged one are indistinguishable to any party holding only the public key. Theorem 1 therefore applies where the signer is honest or otherwise constrained, and it does not apply where the signer is the adversary and benefits from re drawing. Section 6 gives the construction we use in that case.

4.3 Pseudorandomness

Theorem 2. Modelling XOF as a quantum random oracle, for every A issuing q queries there is B , of essentially A 's time, with $\text{Adv}_{\text{PR}}(A) \leq \text{Adv}_{\text{EUF-CMA}}(B) + 2q \cdot 2^{-128}$.

Proof. The image $y^* = \text{XOF}(\sigma^* || x^*)$ is uniform and independent of A 's view unless A queries the oracle at a point $\sigma^* || x^*$, which requires exhibiting σ^* with $\Sigma.Vf(pk, x^*, \sigma^*) = 1$ absent a signature on x^* , an existential forgery that B

extracts by the one way to hiding technique. The additive $2q \cdot 2^{-128}$ bounds the quantum reprogramming distinguishing advantage using the 128 bit quantum preimage margin of a 256 bit output. ■

5 The finality beacon

The protocol finalizes height h by aggregating a two thirds weighted quorum of derandomized ML-DSA attestations into a certificate $C[h]$, guaranteeing certificate soundness. Let $\text{Adv_safety}(t)$ be the maximum advantage of a t time, sub one third adversary in causing two distinct certificates to be accepted at one height.

5.1 Construction

```
seed[0] = H( genesis_nonce )
seed[h] = XOF( seed[h-1] || H(C[h]) || LE64(h) , 256 )
```

The verifier recomputes $\text{seed}[h]$ from public block data. No key, signature, argument, or message is introduced; the sole cost is one XOF over $|\text{seed}[h-1]| + 40$ bytes.

5.2 Bias resistance

Theorem 3. Model XOF as a quantum random oracle. For every A controlling strictly less than one third of weighted stake, running in time t , biasing any bit of $\text{seed}[h]$ with advantage ϵ , there is B in time $t + O(|C|)$ with $\text{Adv_safety}(B) \geq \epsilon - q \cdot 2^{-256}$.

Proof. Fix h . In $\text{seed}[h] = \text{XOF}(\text{seed}[h-1] || H(C[h]) || \text{LE64}(h))$, $\text{seed}[h-1]$ and $\text{LE64}(h)$ are fixed before block h . As XOF is a random oracle, $\text{seed}[h]$ is uniform and independent of A 's view except through the point determined by $H(C[h])$; any advantage exceeding $q \cdot 2^{-256}$ must come from A inducing a certificate $C'[h] \neq C[h]$ accepted at height h . B runs A , observes both certificates, and outputs them as a safety break. By certificate soundness each embeds a two thirds quorum; a sub one third adversary cannot form two disjoint two thirds quorums without an honest double signing, excluded, or an ML-DSA forgery, negligible. Hence the bound. ■

The theorem is conditional on the concrete safety advantage of the deployed protocol, which its own analysis supplies; we establish the reduction. Withholding is handled separately: $\text{seed}[h]$ is a function of the finalized certificate, so declining to finalize forfeits the slot to the liveness path rather than granting a resample, and each finalized block yields exactly one beacon value.

6 Sortition from committed one time preimages

Theorem 1 establishes uniqueness for a signer that derandomizes. It does not establish it against one that does not, and the distinction is not academic. A FIPS 204 signature carries no randomizer, so no verifier holding only the public key can separate a derandomized signature from a hedged one; derandomization is an assumption about the signer, not a rule a protocol can check. Where the drawing party is the adversary the assumption is simply unavailable. In stake weighted committee sortition an adversary hedges, obtains a fresh signature over the same beacon, derives a fresh image, and repeats until the image falls under its threshold. Contemporary ML-DSA signing costs a fraction of a millisecond, so the search is trivial and stake weighting is not weakened but eliminated. We therefore do not use the signature construction for sortition. We give one whose uniqueness is structural.

6.1 Construction

```
Register: R = MerkleRoot( p_0, p_1, ..., p_{L-1} ),   p_i <- $ {0,1}^256
          R is bound to the account's stake bond, and is registered
          before seed[h] is determined for any slot the tree serves.
Draw(N):  d = H( p_N || seed[h] )
          pi = MerkleOpen( R, N, p_N )
Vfy:      MerkleVfy(R, N, p_N, pi) AND d = H( p_N || seed[h] )
Member iff d < T(stake)
```

The verifier rejects an opening at any position other than N , and any root not present in the stake registry. The cost of a draw is one hash and one Merkle opening. There is no signature, no argument of correct evaluation, and no randomness supplied by the drawer.

6.2 Uniqueness is structural

Theorem 4. Model H as a random oracle. For any registered root R and slot index N , no adversary issuing q queries produces two accepted draws (d, π) and (d', π') with $d \neq d'$, except with advantage at most $q \cdot 2^{-128}$.

Proof. An accepted draw exhibits an opening of some preimage at position N against R . Two accepted draws at one (R, N) must either open two distinct preimages at the same position against the same root, which yields a Merkle collision and hence a collision in H , bounded by the birthday term, or open the same p_N while claiming two distinct images $d \neq d'$, contradicting that H is a function. ■

Uniqueness here is by construction rather than by assumption. For a fixed account and slot exactly one image exists, there is no randomizer to vary, and there is nothing to re draw. Because R is committed before the beacon of any slot it serves is determined, and the beacon is unpredictable by Theorem 3, the preimages cannot have been selected for a favourable image. The construction rests on SHA-3 alone.

6.3 Stake splitting, and what it does and does not buy

An adversary may register several bonded accounts, each with its own root, and so obtain several independent draws per slot. Each such draw costs a full stake bond, locked, subject to an unbonding delay, and exposed to slashing. Because the threshold T scales with stake, an adversary splitting stake S across k accounts draws k times against proportionally smaller thresholds, and the expected committee share of the split equals that of S held as a single account. Splitting buys draws, not expected seats, and no draw is free.

One limitation is stated rather than assumed away. Committee membership is stake neutral under splitting because the thresholds scale. Leadership selected by the lowest image is not automatically neutral, since more independent draws afford more opportunities at an extreme value. A stake neutral leader weighting is therefore a design requirement of the deployed protocol, and is established by conformance testing that splitting does not raise measured leadership rather than by assertion. Separately, an account may draw, observe an unfavourable image, and reveal nothing. This forfeits a seat rather than obtaining one, and we record it as an accepted residual rather than a closed case.

7 Quantitative security under quantum attack

We now quantify the work an adversary must expend against each component, distinguishing classical from quantum attack, and show that every component holds above the 128 bit quantum floor that a 256 bit hash establishes. The contrast with the elliptic curve construction QVRF replaces is stark: against ECVRF, a quantum adversary running Shor's algorithm recovers the secret key in a number of operations polynomial in the key length, so its effective work factor collapses toward zero, whereas every QVRF component retains a work factor at or above 2^{128} .

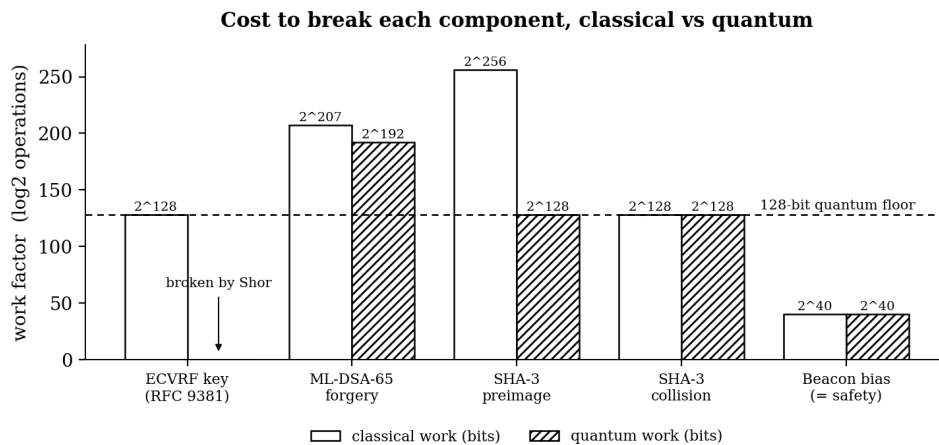


Figure 2. Work factors to defeat each component, in bits, under classical and quantum attack. Shor's algorithm removes the elliptic curve key entirely (leftmost). Grover halves the exponent on hash preimages, leaving 128 bits. Lattice forgery against ML-DSA-65 and the consensus safety bound protecting the beacon both remain above the floor. The dashed line marks the 128 bit quantum target.

6.1 Reading the work factors

Grover's algorithm searches an unstructured space of size N in $\Theta(\sqrt{N})$ quantum operations, so a preimage search against a 256 bit SHA-3 output costs about 2^{128} operations rather than 2^{256} , this is the origin of the 128 bit quantum floor and the reason a 256 bit output, not a 128 bit one, is mandatory. Collision resistance is bounded classically by the birthday value 2^{128} and is not materially improved by quantum search under realistic memory constraints, so we carry 2^{128} for both. ML-DSA-65 targets NIST security category 3, whose forgery resistance is designed to exceed that of a 192 bit block cipher against the best known classical and quantum lattice attacks, no component of which is a Shor style break; we report its margin conservatively at the category level. The beacon carries no cryptographic work factor of its own beyond the negligible $q \cdot 2^{-256}$ collision term, because by Theorem 3 its bias resistance is exactly the safety advantage of the consensus protocol, shown in the figure at a representative 2^{-40} for illustration.

6.2 Composite attack surface

An adversary seeking to subvert QVRF must defeat the weakest necessary component for its goal. To forge a per input output it must break either SUF-CMA of ML-DSA or STARK soundness, each above the floor. To predict a per input output it must break EUF-CMA of ML-DSA or invert XOF, each above the floor. To bias the beacon it must break consensus safety, which is not a cryptographic search at all but a stake threshold, and which no amount of

offline quantum computation reduces. There is no component whose defeat is cheaper under quantum attack than under classical attack, which is the precise sense in which the construction is post quantum by design rather than by retrofit.

8 Parameters and deployment

Sizes correspond to ML-DSA-65, NIST security category 3, taken as the default parameter set.

Object	Size	Frequency
Stored secret (seed)	32 bytes	per key
ML-DSA-65 public key	1,952 bytes	revealed on first proof
ML-DSA-65 signature σ	3,309 bytes	per input, off the finality path
Per input argument π_s	45 to 200 KB	per input, off the finality path
Beacon output seed[h]	32 bytes	per block
Beacon argument	none; C[h] is the witness	per block

Three deployment rules follow. The per input signature and argument never enter consensus messages; they accompany an application transaction, are verified off the finality path, and may be pruned from a data availability layer after verification, so their magnitude does not bear on propagation or finality latency. The per block beacon cost is one XOF over roughly seventy bytes, independent of validator count; before the zero cost property is asserted externally, a benchmark against a beacon free baseline must confirm no measurable difference. Derandomized ML-DSA cannot be enforced by inspection, since the signature does not carry its randomizer; the per input construction is therefore deployed only where the signer is honest or has no incentive to re draw, and the sortition construction of Section 6 is used wherever the drawing party is the adversary. Cryptographic agility is retained by a one byte scheme identifier prefixing every key and signature, so ML-DSA-44, ML-DSA-87, or a future standardized signature is adopted without altering the beacon or the verification interface.

9 Open problems and scope

- Theorems 2 and 3 are in the quantum random oracle model; standard model analogues are open.
- Theorem 3 is conditional on the concrete safety advantage of the deployed protocol.
- The pseudorandomness bound of Theorem 2 is stated in standard, non tightened form.
- Theorem 1 assumes a derandomized signer. That assumption is not verifiable from a FIPS 204 signature, so the per input construction is unsuitable wherever the drawing party is the adversary; Section 6 is used there instead.
- Theorem 4 gives uniqueness per account and slot. Stake splitting remains available at the cost of a bond per draw, and while committee membership is stake neutral under splitting, a stake neutral leader weighting is a requirement on the deployed protocol and is verified by conformance rather than proven here.
- Knowledge soundness is modular, with $\varepsilon_{\text{STARK}}$ from the prover's specification.

This document is released for public cryptanalysis. We characterize QVRF as the first verifiable random function composed solely from NIST standardized post quantum primitives and as a beacon whose bias resistance reduces to consensus safety; both are supported by the reductions herein under their stated models. No result is represented as unconditionally secure.

10 References

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